

BCSC0011: THEORY OF AUTOMATA & FORMALLANGUAGES

Objective: The objective of this course is that students will study and compare different models and views of the abstract notion of computation and its various aspects.

Credits:04

Semester V

L-T-P-J:3-1-0-0

Module No.	Content	Teaching Hours
I	Introduction: Alphabets, Strings and Languages; Automata and Grammars, Deterministic Finite Automata (DFA), Nondeterministic Finite Automata (NFA), Equivalence of NFA and DFA, Minimization of Finite Automata, Myhill-Nerode Theorem; FA with Output - Moore and Mealy machine, Applications and Limitations of FA. Regular expression (RE): Regular Expression to FA, DFA to Regular Expression, Arden Theorem, Non Regular Languages, Pumping Lemma for Regular Languages, Applications of Pumping Lemma, Closure Properties of Regular Languages. Push Down Automata (PDA): Introduction, Language of PDA, Acceptance by Final State, Acceptance by Empty Stack, Deterministic PDA.	20
II	Context Free Grammar (CFG) and Context Free Languages (CFL): Introduction, Derivation Trees, Ambiguity in Grammar, Ambiguous to Unambiguous CFG, Simplification of CFGs, Normal Forms for CFGs - CNF and GNF; Pumping lemma for CFLs, Equivalence of PDA and CFG. Turing machines (TM): Basic Model, Definition and Representation, Variants of Turing Machine and their equivalence, TM for Computing Integer Functions, Universal TM, Church's Thesis, Recursive and Recursively Enumerable Languages, Halting Problem, Introduction to Computational Complexity.	20

Text Books:

- K.L.P. Mishra and N. Chandrasekaran, "Theory of Computer Science: Automata, Languages and Computation", 3rd Edition, PHI,2006

Reference Books:

- Hopcroft, Ullman , "Introduction to Automata Theory, Languages and Computation", 3rd Edition, Pearson Education,2013.
- Martin J. C.," Introduction to Languages and Theory of Computations", 4th Edition, TMH,2011.

Outcome: After completion of course, the student will be able to:

- C01: Understand the basic concepts of Context Free languages, Expression and Grammars.
- C02: Analyze the conversion of NFA to DFA, Mealy to Moore and Moore to Mealy.
- C03: Analyze the process to convert regular expression to DFA, DFA to regular expression, and minimization of DFA.
- C04: Develop the PDA for the context free language and context free grammar.
- C05: Analyze that the grammar is ambiguous or unambiguous.
- C06: Apply the process to convert CFG to CNF and GNF.
- C07: Understand the concept of Turing machine and its variants.
- C08: Design the Turing machine for the real world application.